American Journal of Emergency Medicine 104 (2026) 95–101

Contents lists available at ScienceDirect

American Journal of Emergency Medicine

journal homepage: www.elsevier.com/locate/ajem

Automated auditing of emergency department documentation using large language models

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ARTICLE INFO

Article history:
Received 1 January 2026
Received in revised form 20 February 2026
Accepted 10 March 2026

Keywords:
Artificial intelligence
Emergency department documentation
Large language models
Patient safety
Quality assurance

ABSTRACT

Introduction: Errors in emergency department (ED) documentation can lead to patient harm and medicolegal risk, however manual document auditing is resource-intensive and difficult to scale. Large language models (LLMs) may offer an automated alternative, but their performance for task-specific error detection remains uncertain. This study evaluated four state-of-the-art LLMs for automated auditing of ED documentation.
Methods: A total of 600 ED discharge notes from real-world encounters were evaluated, including 500 notes with a single injected documentation error across five predefined categories (patient/encounter, medication/allergy, clinical findings, investigation failure/contradiction, or disposition/follow-up), and 100 error-free reference notes. Four LLMs (Claude Sonnet 4.5, Gemini 2.5 Pro, GPT-5, and Grok 4) independently audited each note under blinded conditions. Model performance was assessed using accuracy, precision, recall, and F1 score, with pairwise comparisons performed using McNemar's test.
Results: Model accuracies ranged from 93.2% to 94.2% for detection of any documentation error, with precision exceeding 98% and F1 scores above 0.95 for all models. Claude Sonnet 4.5 achieved the highest overall accuracy (94.2%) and F1 score (0.964). Pairwise comparisons showed no statistically significant differences in overall error detection between models ($p > 0.05$). At the category level, all models demonstrated near-perfect performance for patient/encounter inconsistencies and medication/allergy errors (F1 scores >0.96), with lower performance for investigation failures/contradictions, followed by disposition/follow-up errors and clinical finding inconsistencies.
Conclusion: LLMs demonstrate strong performance in auditing ED documentation, supporting their potential role as quality assurance tools in emergency care. Future work should explore how these systems can be integrated into real clinical workflows.
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1. Introduction

The emergency department (ED) discharge process represents a high-risk transition of care where communication failures can lead to significant patient harm [1,2]. Electronic health record documentation errors, which range from contradictory discharge instructions to missed critical laboratory results, are a recognized source of medicolegal liability and adverse outcomes [3-6]. While physician peer review and manual chart auditing are currently the gold standards for quality assurance [4,7,8], these processes are resource-intensive, difficult to scale, and often retrospective, occurring too late to mitigate immediate patient risk.

Recent advances in large language models (LLMs) suggest that these systems may have the semantic reasoning capabilities necessary to identify clinically meaningful inconsistencies within unstructured clinical text [9,10]. While prior studies have evaluated LLMs for tasks such as clinical summarization, report generation, or diagnostic assistance, comparatively less work has examined their potential role as automated auditors for discharge documentation safety, particularly using explicit reference standards for documentation failures [11-15]. In addition, few studies have reported model inference run time in clinically relevant settings [9,16]. As a result, the feasibility of deploying these models as assistive quality assurance tools in emergency medicine remains uncertain.

To address this gap, we evaluated the performance of four state-of-the-art LLMs in identifying ED discharge documentation errors. Using discharge notes derived from real-world ED encounters, this study aimed to assess the potential utility and limitations of LLMs as scalable tools for discharge documentation quality assurance in emergency care.

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https://doi.org/10.1016/j.ajem.2026.03.007
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2. Methods

2.1. Database

This study used data from the MIMIC-IV-ED database [17,18], a large, single-institution, de-identified repository of ED encounters. From this database, 600 ED visits were randomly selected to form the study cohort. For each selected encounter, structured clinical data elements, including patient demographics, triage information, documented vital signs, diagnoses, administered medications, and disposition, were extracted and used to derive standardized ED documentation for downstream analysis.

2.2. Error categories and injection

To establish a ground-truth reference standard, five predefined categories of ED documentation errors were systematically injected into 500 discharge notes [10], with each note containing only one error (Fig. 1). These categories were defined a priori to represent common, clinically meaningful documentation failure modes in electronic health

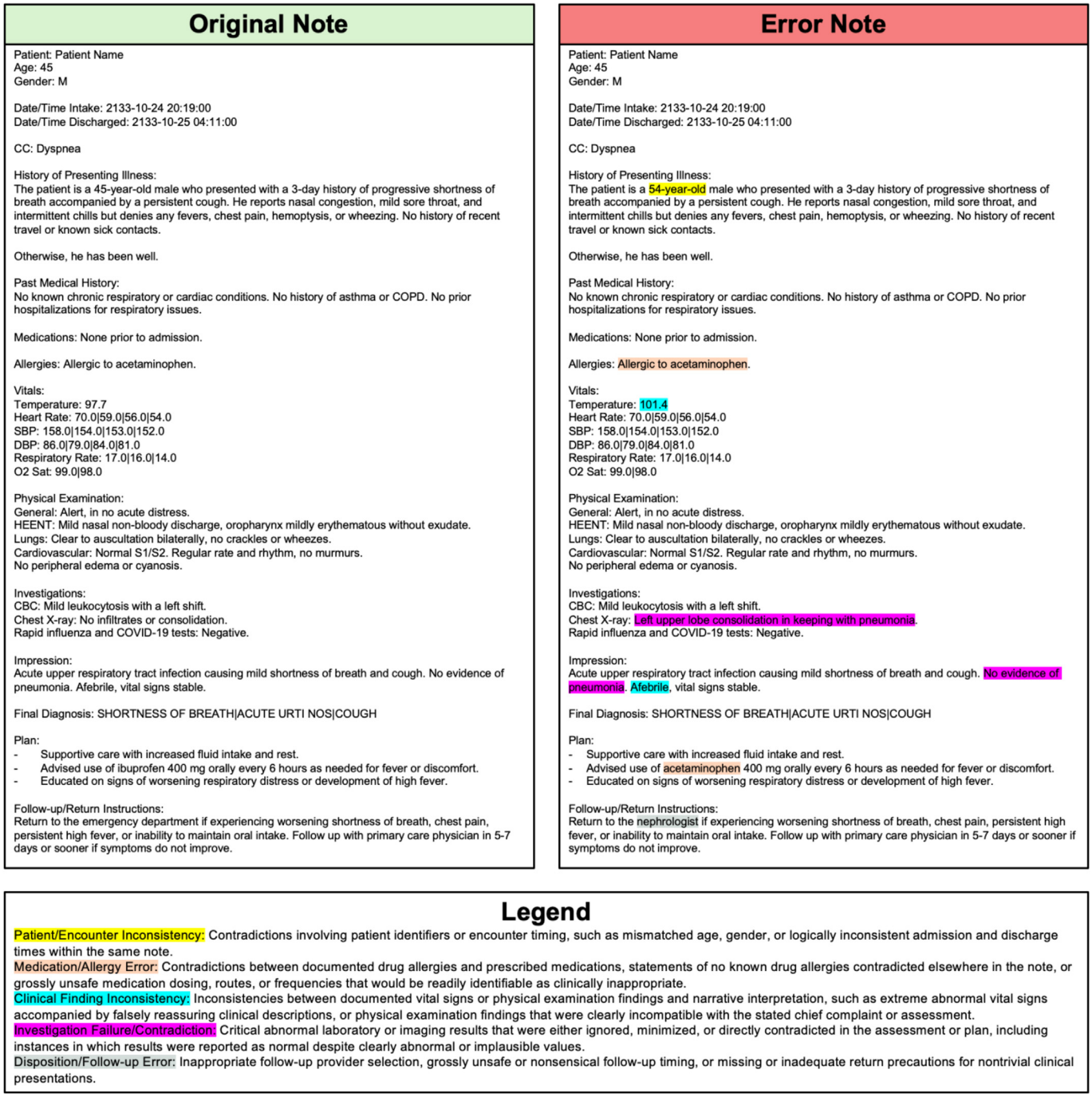


Fig. 1. Representative example illustrating the predefined emergency department discharge documentation error categories. The original discharge note is shown on the left, and a corresponding error-injected note is shown on the right. For illustrative purposes, the error note highlights examples from multiple error categories; however, in the study dataset, each discharge note was injected with only a single predefined error category to preserve a clear ground-truth reference standard.

records (EHRs) and ED discharge communication, informed by prior quality and patient safety literature [5,6,19-22]. The final dataset contained 100 discharge notes for each error category and 100 error-free control notes.

The first category, patient/encounter inconsistency, consisted of patient demographic or temporal inconsistencies, defined as internal contradictions involving patient identifiers or encounter timing, such as mismatched age, gender, or logically inconsistent admission and discharge times within the same note. The second category, medication/allergy errors, included medication or allergy safety errors, including explicit contradictions between documented drug allergies and prescribed medications, statements of no known drug allergies contradicted elsewhere in the note, or grossly unsafe medication dosing, routes, or frequencies that would be readily identifiable as clinically inappropriate. The third category, clinical finding inconsistency, included gross inconsistencies between documented vital signs or physical examination findings and narrative interpretation, such as extreme abnormal vital signs accompanied by falsely reassuring clinical descriptions, or physical examination findings that were clearly incompatible with the stated chief complaint or assessment. The fourth category, investigation failure/contradiction, captured investigation failures or contradictions, defined as critical abnormal laboratory or imaging results that were either ignored, minimized, or directly contradicted in the assessment or plan, including instances in which results were reported as normal despite clearly abnormal or implausible values. The fifth category, disposition/follow-up errors, consisted of disposition, follow-up, or return precaution errors, including inappropriate follow-up provider selection, grossly unsafe or nonsensical follow-up timing, or missing or inadequate return precautions for nontrivial clinical presentations.

Error injection and category assignment were performed by two human reviewers with formal medical training following a standardized approach designed to preserve clinical plausibility, maintain the original discharge note structure, and avoid the introduction of unintended errors. Each discharge note was initially modified by a single reviewer, then independently reviewed by a second reviewer to verify error category assignment, clinical plausibility, and absence of unintended secondary errors. Disagreements were resolved through structured discussion to consensus. Fig. 2 provides an overview of the study workflow for automated ED document auditing.

2.3. Large language models

Four contemporary LLMs from distinct vendors were evaluated as automated auditors: GPT-5 (OpenAI), Gemini 2.5 Pro (Google DeepMind), Claude Sonnet 4.5 (Anthropic), and Grok 4 (xAI). Each model was used exclusively in an auditing role and did not participate in discharge note derivation or error assignment. All models received an identical system prompt that defined the five predefined documentation error categories and specified explicit criteria for positive and negative classification (Supplemental Fig. S1). Using their respective application programming interface (API), all models were instructed to independently review each discharge note and to label the presence or absence of each error category. Auditing was conducted under group-blinded conditions, such that models had no access to encounter metadata, file structure, or error assignments. Inference was performed with the temperature parameter set to zero, when applicable, to minimize stochastic variability and support reproducibility of model outputs. Additional model details are provided in Supplemental Table S1.

2.4. Statistical analysis

Statistical analyses compared model performance at both the overall level and within individual documentation error categories. For each model, overall performance was first evaluated for the detection of any documentation error, defined as a binary outcome combining all categories. Overall detection performance was then compared pairwise between models using McNemar's test on matched discharge notes. Within-model comparisons across error categories were also performed

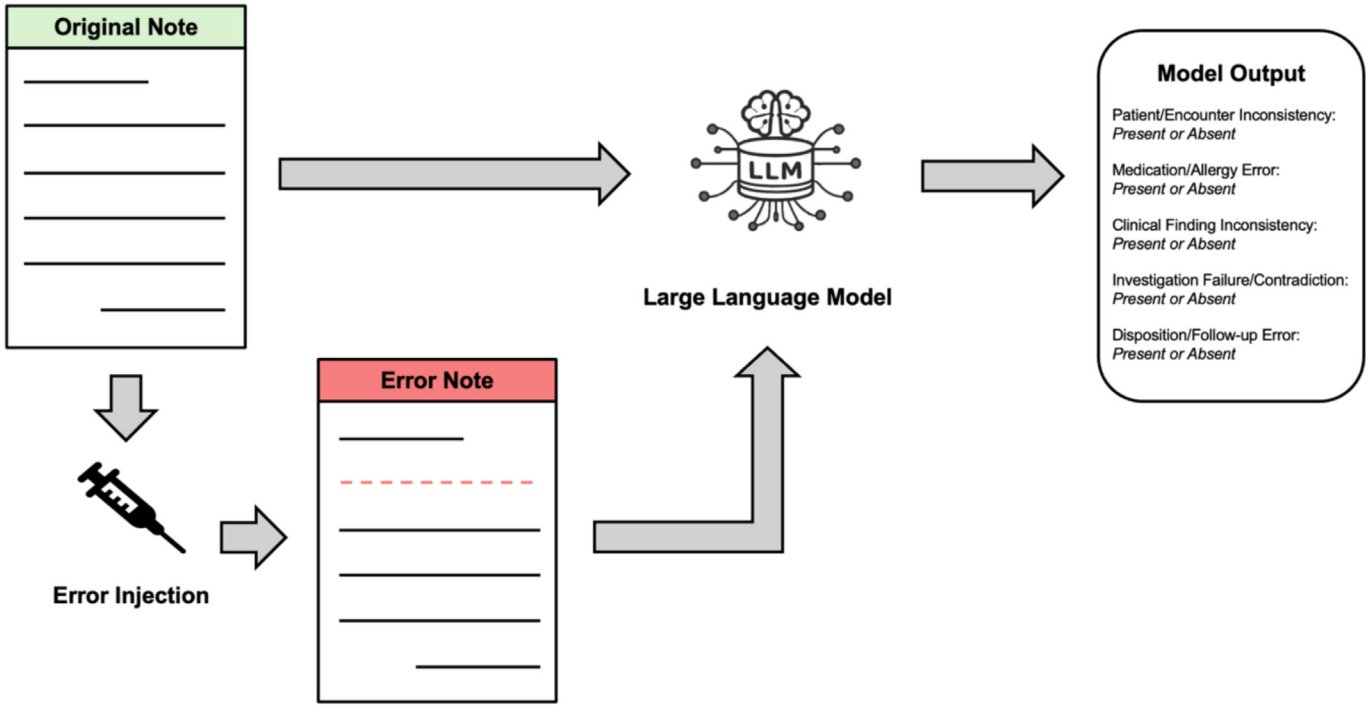


Fig. 2. Overview of the study workflow for automated auditing of emergency department discharge documentation. Original discharge notes were either evaluated unchanged or systematically modified through controlled error injection to create error notes containing a predefined documentation error. Both original and error-injected notes were independently reviewed by a large language model (LLM), which classified the presence or absence of five documentation error categories: patient/encounter inconsistency, medication/allergy error, clinical finding inconsistency, investigation failure/contradiction, and disposition/follow-up error.

to assess category-dependent variability in performance. In addition, between model, within documentation error category performance was also compared using McNemar's test.

To account for multiple comparisons, the Holm–Bonferroni method was applied separately within the three predefined families of hypothesis tests: overall pairwise model comparisons for any-error detection, within-model pairwise comparisons across documentation error categories, and between-model pairwise comparisons within each documentation error category. This family-wise correction strategy was selected to control the type I error rate within each analytical domain while preserving statistical power across related comparisons.

Performance metrics included accuracy, precision (positive predictive value), recall (sensitivity), specificity, negative predictive value, and F1 score. In addition, inference characteristics were recorded to support implementation considerations. Inference time was measured as wall-clock latency from prompt submission to receipt of the complete model response under stable network conditions. Input, output, and total token counts were obtained from API usage metadata for each model call. Statistical significance was defined as a two-sided p-value <0.05.

3. Results

3.1. Overall model performance for any error category

All four LLMs demonstrated high performance in identifying any document error (Table 1). Overall classification accuracy ranged from 93.2% to 94.2% across models, with Claude Sonnet 4.5 achieving the highest accuracy (94.2%), followed by Gemini 2.5 (94.0%), Grok 4 (93.5%), and GPT-5 (93.2%). Precision for identifying the presence of any documentation error was uniformly high across all LLMs, ranging from 98.3% to 99.1%. Grok 4 demonstrated the highest precision (99.1%), followed by GPT-5 (98.9%), Gemini 2.5 (98.7%), and Claude Sonnet 4.5 (98.3%). Recall was highest using Claude Sonnet 4.5 (94.6%), followed by Gemini 2.5 (94.0%), Grok 4 (93.0%), and GPT-5 (92.8%). Specificity ranged from 92.0% to 96.0% across models, while NPV was comparatively lower, ranging from 72.5 to 77.3%. Collectively, all four LLMs yielded F1-scores ranging from 0.958 to 0.964, with Claude Sonnet 4.5 achieving the highest aggregate F1 score (0.964).

Pairwise model comparisons using McNemar's test demonstrated no statistically significant differences in overall performance between models following adjustment for multiple comparisons (adjusted $p > 0.05$ for all model pairs). Supplemental Table S2 provides a macro-averaged summary of performance across models.

3.2. Within-model performance across error categories

Across all four LLMs, category-level performance demonstrated within-model heterogeneity (Table 2). Patient/encounter inconsistencies and medication/allergy errors were consistently among the highest-performing categories across models, with F1 scores typically exceeding 0.95. Clinical finding inconsistencies were the lowest-performing category, with F1 scores ranging from 0.75 to 0.84 (Fig. 3), driven primarily by reduced recall.

Within-model McNemar testing confirmed that, for GPT-5, Claude Sonnet 4.5, Gemini 2.5, and Grok 4, clinical finding inconsistencies were detected significantly less frequently than patient/encounter inconsistencies and medication/allergy errors after adjustment ($p < 0.05$ for all models). For GPT-5 and Claude Sonnet 4.5, clinical finding inconsistencies were also detected significantly less frequently than investigation failures/contradictions, while no statistically significant difference was observed between clinical finding inconsistencies and disposition/follow-up errors.

Moreover, for Gemini 2.5 Pro and Grok 4, disposition/follow-up errors were detected significantly more frequently than investigation failures/contradictions, whereas no significant difference between these categories was observed for GPT-5 or Claude Sonnet 4.5. Supplemental Table S3 summarizes each pairwise McNemar comparison within models across each documentation error category.

3.3. Between-model performance within error categories

Between-model performance was largely comparable for patient/encounter inconsistencies and medication/allergy errors, with all four models demonstrating near-ceiling performance in these categories (F1 scores ≥0.96). Pairwise McNemar comparisons showed no statistically significant differences between models for patient/encounter inconsistencies or medication/allergy errors after adjustment, indicating broadly equivalent performance in these structured documentation domains.

Greater between-model differences were observed for clinical finding inconsistencies, which represented the most challenging category across systems. Grok 4 demonstrated the highest performance in this domain (F1 score = 0.840; accuracy = 95.0%) and significantly outperformed GPT-5 after adjustment (adjusted $p = 0.048$), reflecting fewer discordant misclassifications on paired discharge documents. No other pairwise comparisons for clinical finding inconsistencies were statistically significance after correction.

For investigation failures/contradictions, performance differences between models were modest. Although Grok 4 and Claude Sonnet 4.5 achieved numerically higher F1 scores than GPT-5 and Gemini 2.5 Pro, none of the pairwise comparisons remained statistically significant after adjustment, indicating no clear between-model superiority in this category.

The largest and most consistent between-model differences were observed for disposition/follow-up errors. Claude Sonnet 4.5 achieved the highest performance in this domain (F1 score = 0.957; sensitivity = 0.99), significantly outperforming Gemini 2.5 Pro and Grok 4 after Bonferroni adjustment (adjusted $p < 0.01$ for both comparisons). Gemini 2.5 Pro demonstrated the lowest performance in this category (F1 score = 0.793), driven primarily by reduced precision (0.686). Supplemental Table S4 summarizes the pairwise McNemar comparisons between models within each documentation error category.

3.4. Inference time and token utilization

Median inference time and token utilization varied across models. Claude Sonnet 4.5 had a median inference time of 4.1 s (interquartile range [IQR] 3.1–5.1), with 1374 input tokens (IQR 1322–1467), 230 output tokens (IQR 193–265), and a total of 1623 tokens (IQR 1547–1697). Gemini 2.5 Pro had a median inference time of 11.4 s (IQR 9.7–14.4), with 1253 input tokens (IQR 1212–1324), 1220 output tokens (IQR

Table 1
Overall performance of large language models for detection of any documentation error. Abbreviations: PPV = positive predictive value, NPV = negative predictive value.

Model	Accuracy	Precision (PPV)	Recall (Sensitivity)	Specificity	NPV	F1 Score
Claude Sonnet 4.5	0.942	0.983	0.946	0.920	0.773	0.964
Gemini 2.5 Pro	0.940	0.987	0.940	0.940	0.758	0.963
GPT-5	0.932	0.989	0.928	0.950	0.725	0.958
Grok 4	0.935	0.991	0.930	0.960	0.733	0.960

Table 2
Category-level performance of large language models across documentation error categories. Abbreviations: PPV = positive predictive value, NPV = negative predictive value. Note: Metrics are reported across $n = 600$ notes per category (100 positive).

Model	Error Category	Accuracy	Precision (PPV)	Recall (Sensitivity)	Specificity	NPV	F1
Claude Sonnet 4.5	Patient/Encounter Inconsistency	0.988	0.960	0.970	0.992	0.994	0.965
	Medication/Allergy Error	0.987	0.951	0.970	0.990	0.994	0.960
	Clinical Finding Inconsistency	0.940	0.840	0.790	0.970	0.958	0.814
	Investigation Failure/Contradiction	0.980	0.907	0.980	0.980	0.996	0.942
	Disposition/Follow-up Error	0.985	0.925	0.990	0.984	0.998	0.957
Gemini 2.5 Pro	Patient/Encounter Inconsistency	0.997	0.990	0.990	0.998	0.998	0.990
	Medication/Allergy Error	0.993	0.980	0.980	0.996	0.996	0.980
	Clinical Finding Inconsistency	0.932	0.804	0.780	0.962	0.956	0.792
	Investigation Failure/Contradiction	0.970	0.853	0.990	0.966	0.998	0.917
	Disposition/Follow-up Error	0.918	0.686	0.940	0.914	0.987	0.793
GPT-5	Patient/Encounter Inconsistency	0.997	0.990	0.990	0.998	0.998	0.990
	Medication/Allergy Error	0.998	1.000	0.990	1.000	0.998	0.995
	Clinical Finding Inconsistency	0.927	0.841	0.690	0.974	0.940	0.758
	Investigation Failure/Contradiction	0.968	0.865	0.960	0.970	0.992	0.910
	Disposition/Follow-up Error	0.942	0.756	0.960	0.938	0.992	0.846
Grok 4	Patient/Encounter Inconsistency	0.992	0.961	0.990	0.992	0.998	0.975
	Medication/Allergy Error	0.990	0.961	0.980	0.992	0.996	0.970
	Clinical Finding Inconsistency	0.950	0.898	0.790	0.982	0.959	0.840
	Investigation Failure/Contradiction	0.985	0.950	0.960	0.990	0.992	0.955
	Disposition/Follow-up Error	0.957	0.856	0.890	0.970	0.978	0.873

1066–1494), and a total of 2492 tokens (IQR 2307–2772). GPT-5 had a median inference time of 22.9 s (IQR 17.9–31.1), with 1174 input tokens (IQR 1137–1250), 1224 output tokens (IQR 926–1662), and a total of 2454 tokens (IQR 2115–2841). Grok 4 also had a median inference time of 22.9 s (IQR 19.3–29.0), with 1812 input tokens (IQR 1772–1875), 899 output tokens (IQR 753–1089), and a total of 2715 tokens (IQR 2537–2979).

4. Discussion

To our knowledge, this study represents the first evaluation of LLMs as automated auditors for ED discharge documentation. Using notes derived from real-world data with carefully injected, clinically meaningful documentation errors, we demonstrate that contemporary LLMs can reliably detect a broad range of discharge documentation errors with strong performance across all evaluated domains.

These findings extend prior work evaluating LLMs for clinical summarization, diagnostic reasoning, and guideline adherence by demonstrating feasibility in safety-focused auditing of clinical text [9,11,19,23,24]. For example, a recent study by Chung et al. evaluated LLM-based fact verification of clinical documents generated by LLMs using patient EHRs as ground truth and reported 93.2% agreement with clinician review [24]. While that work examined the factual accuracy of generated narratives, our study evaluates a related but distinct task involving identification of predefined, clinically meaningful documentation errors in real-world ED discharge notes. Within this structured auditing framework, we observed overall detection accuracy within a comparable range, supporting the feasibility of LLMs for medical documentation auditing applications [24].

At the error-category level, performance variability across models provides insight into both the strengths and limitations of current LLM architectures. All models demonstrated near-perfect performance for

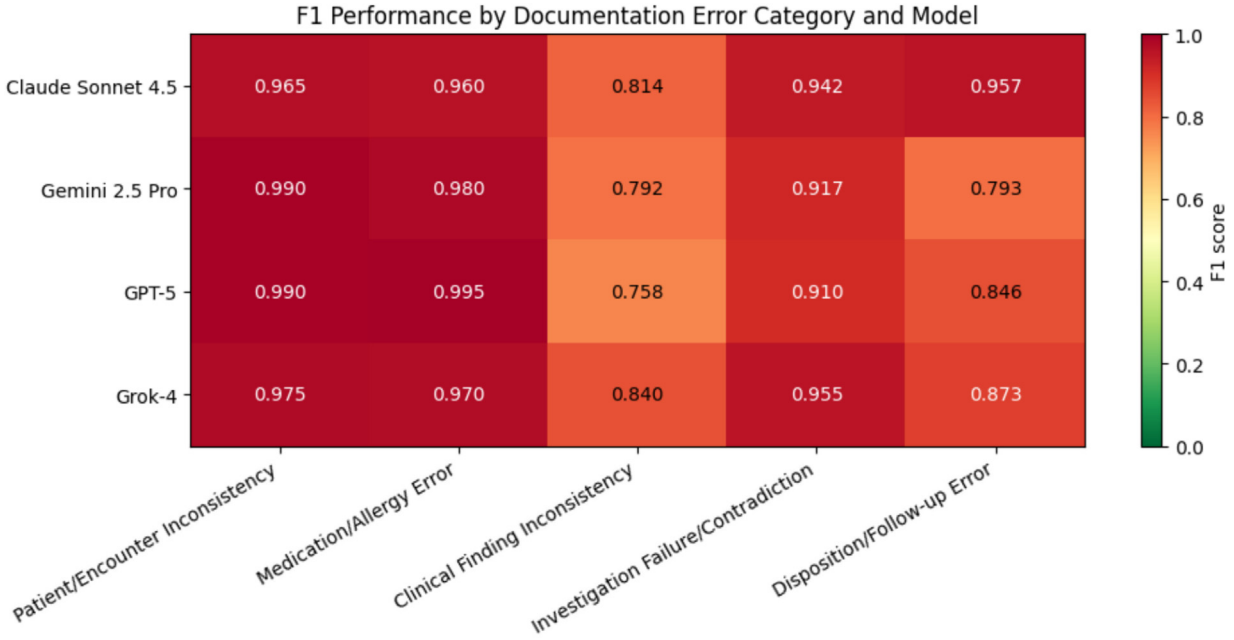


Fig. 3. F1 scores by documentation error category across evaluated large language models.

patient/encounter and medication or allergy-related errors, reflecting the relative clarity and explicitness of these error types, with investigation failures/contradictions also detected with high accuracy. In contrast, clinical finding inconsistencies and disposition/follow-up errors were more challenging, particularly for GPT-5 and Gemini 2.5 Pro. These errors may require more nuanced clinical judgment and contextual interpretation, which may not be fully captured by general-purpose language models. Claude Sonnet 4.5 and Grok 4 consistently outperformed other models in these categories, suggesting that differences in model training strategies or internal reasoning mechanisms may meaningfully affect performance, although the proprietary nature of these systems limits attribution to specific architectural features [25-27].

From a clinical and workflow perspective, these findings suggest that LLMs may hold promise as assistive auditing tools for discharge documentation quality assurance. The overall detection accuracy observed in this study falls within performance levels reported for structured human chart auditing tasks, although direct head-to-head comparison was not performed. Prior quality-assurance studies indicate that human performance on structured medical chart auditing tasks is variable, with reported accuracy or agreement commonly ranging from approximately 70 to 95% [28,29], depending on reviewer training, experience, and task definition. In addition to variability in agreement, manual auditing requires dedicated clinician or trainee time and formal training, which may limit scalability in high-volume emergency department settings [30]. In contrast, LLM-based auditing in this study was completed with median wall-clock inference times on the order of seconds per note, whereas manual chart review can take minutes to hours, depending on documentation complexity and reviewer experience [30,31].

Within this framework, high specificity and consistently high per-category NPV across models indicate a low risk of false positive flagging at the individual error level, which may help maintain clinician trust and minimize alert fatigue. However, when errors are aggregated for global any-error detection, NPV is inherently lower, reflecting the increased probability that at least one documentation issue is present across multiple assessed domains. These findings suggest that real-world use would likely require a clearly defined operational role. For example, documentation domains with greater anticipated clinical impact might be subject to more conservative thresholds or prioritized human review, whereas lower-risk categories could be reviewed selectively or in aggregate. In such a model, automated auditing could function within a human-in-the-loop workflow that helps triage documentation for clinician oversight based on predefined criteria such as error category, confidence of detection, or the presence of multiple concurrent flags. Clinical oversight would remain central, with automated tools serving to surface potential concerns rather than to completely replace human judgment, and implementation would require careful calibration of thresholds and prospective evaluation to balance efficiency with patient care [32,33].

Future research should explore strategies to improve model performance. Approaches such as fine-tuning or instruction-tuning models on domain specific discharge documentation corpora may improve sensitivity for nuanced or rare error types, while more advanced prompt engineering or agentic auditing frameworks incorporating guideline retrieval and cross document consistency checks may further enhance performance [34-38]. In parallel, prospective evaluation of these systems as live, point-of-care auditing tools will be necessary to assess real-time utility, workflow integration, clinician acceptance, and their ability to identify documentation vulnerabilities before patient discharge.

Several broader considerations also warrant attention before clinical deployment. Regulatory oversight, medicolegal liability, and ethical accountability for AI-assisted documentation auditing remain unresolved, and to date no LLM has been approved by the FDA or Health Canada for standalone clinical use [39-41]. Accordingly, transparency regarding model limitations, delineation of responsibility between clinicians and automated systems, and safeguards against automation bias will be essential as these tools move toward potential clinical adoption [42-44]. In addition, observed variability in performance across documentation error categories raises important equity considerations if certain types of documentation failures are systematically more likely to be missed [42,43,45].

This study is not without limitations. Documentation errors were systematically injected into discharge notes rather than occurring naturally, which may not fully capture the complexity of real-world documentation errors. However, this controlled approach enabled the use of a clear reference standard and allowed for paired, category-specific performance comparisons across models. In addition, the dataset was derived from a single institution, which may limit generalizability to other ED settings with different documentation practices or patient populations. Inference time and token utilization estimates reported in this study reflect observed performance under specific API conditions and may vary with server load, network latency, model hosting environment, and prompt length. Finally, models were evaluated in a static inference setting using uniform and consistent prompts delivered through vendor APIs. Performance may differ under alternative prompting strategies, inference configurations, or as model versions evolve over time [34,35].

5. Conclusion

Our findings demonstrate that LLMs can accurately identify many clinically meaningful discharge documentation errors, positioning them as potential tools to support quality assurance processes in emergency medicine. Future work should explore prospective clinical evaluation, assessment of additional error types, and safe integration into real-world workflows.

CRediT authorship contribution statement

Nicholas Dietrich: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Brett Stubbert:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajem.2026.03.007.

Data availability

The study framework code and system prompts used for large language model inference are available from the corresponding author upon reasonable request.

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